

EXPERIMENT C₈

HEAT OF SOLUTION FROM MEASUREMENTS OF SOLUBILITY

Introduction:

One of the best known and most useful thermodynamic formulae is the van't Hoff equation, which relates the equilibrium constant of a reaction, K , to the standard enthalpy change of that reaction, ΔH° . It is:

$$\frac{d \ln K^\circ}{dT} = \frac{\Delta H^\circ}{RT^2}$$

A similar equation can be derived relating the solubility of a solid to its enthalpy of solution:

$$\frac{d \ln (S/S^\circ)}{dT} = \frac{\Delta H^\circ}{RT^2}$$

where S is the solubility in moles per 1000 g of solvent, and ΔH° the standard enthalpy of solution, i.e. the enthalpy change for the solution of benzoic acid in water



and $S^\circ = 1 \text{ mol kg}^{-1}$. Assuming ΔH° to be constant over the temperature range considered, we get, on integration,

$$\ln (S/S^\circ) = -\frac{\Delta H^\circ}{RT} + c$$

Thus on plotting $\ln (S/S^\circ)$ against T^{-1} we should get a straight line of slope $-\Delta H^\circ/R$, from which ΔH° can be found.

Procedure:

The thermostat should be set at about 40 °C when you enter the laboratory. A bottle containing a saturated solution of benzoic acid and excess solid benzoic acid will be supplied. Shake the bottle and transfer about 150 cm³ to the conical flask supplied. The solution should contain a few grams of undissolved benzoic acid. Place the flask in a beaker of water which has been heated to about 50 °C and leave it there until it has reached that temperature, keeping the solution well stirred. If the solid disappears more solid benzoic acid should be added. Transfer the flask to the thermostat and again wait until temperature equilibrium has been reached. Stir from time to time. In the meantime weigh a 100 cm³ conical flask. When the solution is at the

correct temperature transfer about 10 cm^3 to this flask using the glass wool trap provided. This is to prevent small particles of solid benzoic acid from entering the pipette.

To avoid crystals separating out in the pipette, heat the pipette *carefully* in a bunsen flame before use. This can be done because the pipette is not being used to deliver a fixed volume. Weigh the flask plus solution and titrate the benzoic acid against the $0.015 \text{ mol dm}^{-3}$ NaOH supplied, using phenolphthalein as the indicator.

Leave the 200 cm^3 flask in the thermostat and drop the temperature to about 35°C . Repeat the above procedure and do further determinations at 30°C , 25°C and 20°C . Calculate the enthalpy of solution as explained above.

Notes:

- (i) In this experiment the solution is cooled from a higher temperature because equilibrium between the solid and the solution is established relatively rapidly in this case. The presence of the solid prevents super saturation. The establishment of equilibrium when the solid and solution are heated is, however, a very slow process and would take many hours.
- (ii) Temperatures should be measured as accurately as possible using the thermometers provided.

Question:

Is the sign of ΔH° in accord with Le Chatelier's principle? Discuss.